Lynbrook Math Summer Camp 2026
A1 Algebra Manipulations

Lynbrook Math Club

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A1.1 Lecture

The world is full of unknowns. Algebra is the branch of mathematics that covers the roles of expressions and unknown variables in an algebraic system. In this lesson, we will cover clever ways of solving for unknowns in equations.

Evaluation of Expressions with the Order of Operations

Before we start solving any fancy equations, we must first be able to evaluate expressions. An expression is a string of numbers, variables, and operations. Rather than evaluating the expression from left to right, the operations must be performed in a specific order to arrive at the correct answer. This order of operations is often known as PEMDAS.

Parentheses

The numbers, variables, and operations within a set of parentheses () should be treated as an expression within the overall expression that should be evaluated first. The expression within the parentheses should also be evaluated with the correct order of operations.

Example 1
Evaluate $7 \times (3 + 1) \times 5$.

We first evaluate the expression within the parentheses, then multiply the result with the other numbers:

$$7 \times (3 + 1) \times 5 = 7 \times 4 \times 5 = 140.$$

Exponents and Other Functions

Next, evaluate exponents and any other functions present within the expression. This includes square roots, factorials, absolute values, logarithms, floor and ceiling, etc.

Example 2
Evaluate $16 \div 2^4 + |10 - 4|$.

There are no parentheses, so we first evaluate the functions. In this case, they are 2^4 and $|10-4|$.

$2^4 = 16$, and to evaluate $|10-4|$, we would first evaluate the expression within the absolute value signs, and then apply the absolute value function. In this way, we treat whatever is within the function as if it had parentheses around it. This works the same way with other functions.

The expression evaluates as:

$$16 \div 2^4 + |10-4| = 16 \div 16 + |-6| = 1 - 6 = -5.$$

Multiplication and Division

Apply the multiplication and division operations next. Multiplication follows the commutative, associative, and distributive properties.

Claim A1.1.3 (Commutative Property) — The commutative property states for a given commutative operation, $a\#b = b\#a$.

Claim A1.1.4 (Associative Property) — The associative property states that for a given associative operation $\#$, $a\#(b\#c) = (a\#b)\#c$.

Claim A1.1.5 (Distributive Property) — The distributive property for multiplication states that $a \times (b+c) = ab + ac$.

Division is not commutative nor is it associative. Also note that division by zero is undefined. However, division is merely multiplication by the reciprocal of the divisor.

Definition A1.1.6 (Reciprocal). The reciprocal of an expression x is $\frac{1}{x}$.

Since division can be rewritten as multiplication, we can now apply the commutative and associative properties.

Example 7

Simplify $2 \times 4 \times 9 \times 7 \times 5 \div 2$.

We can rewrite the expression as $2 \times 4 \times 9 \times 7 \times 5 \times \frac{1}{2}$, and then apply the commutative property:

$$2 \times 4 \times 9 \times 7 \times 5 \times \frac{1}{2} = 2 \times \frac{1}{2} \times 4 \times 5 \times 9 \times 7 = 1 \times 20 \times 63 = 126.$$

Addition and Subtraction

Addition and subtraction can be evaluated similarly to multiplication and division, with addition following the commutative and associative properties, and subtraction can be rewritten as addition of the negative of the subtrahend.

Simplification Techniques

Mathematicians are lazy, so we try to make our expressions and equations as short and simple as possible. There are many techniques for this.

Exponent and Square Root Rules

Exponents are repeated multiplication, so we can derive the following:

$$x^a \cdot x^b = x^{a+b} \tag{1}$$
$$x^a \div x^b = x^{a-b} \tag{2}$$
$$(x^a)^b = x^{ab} \tag{3}$$
$$\frac{1}{x^a} = x^{-a} \tag{4}$$
$$\sqrt[a]{x^b} = x^{\frac{b}{a}} \tag{5}$$

Common Denominators

We cannot add or subtract fractions with unlike denominators, so we must manipulate the fractions into a form such that they have the same denominators. We do this by multiplying the numerators and denominators of one fraction by the denominator of the other fraction, and vice versa.

Example 8

Simply $\frac{13}{5} + \frac{9}{4}$.

We multiply the first fraction by $\frac{4}{4}$ and the second fraction by $\frac{5}{5}$ to achieve a common denominator. This results in:

$$\frac{12}{5} \cdot \frac{4}{4} + \frac{9}{4} \cdot \frac{5}{5} = \frac{48}{20} + \frac{45}{20} = \frac{93}{20}.$$

Cross Multiplication

Cross multiplication is a way of dealing with isolated fractions on both sides of an equation:

Claim A1.1.9 (Cross Multiplication) — Given $\frac{a}{b} = \frac{c}{d}$, $ab = bc$ given that $b \neq 0$ and $d \neq 0$.

This is possible because we multiply both sides of the equation by bd.

Example 10

Solve for x: $\frac{4}{x+3} = \frac{1}{x+6}$.

x is in the denominator, which we do not like, so we cross multiply:

$$4(x+6) = 1(x+5) \Rightarrow 4x + 24 = x + 3.$$

We can then solve the equation from here.

Rationalizing Denominators

When working with fractions, mathematicians prefer not to have square roots in the denominators. We rationalize the denominator of a fraction by multiplying the numerator and denominator by the square root that is in the denominator.

Example 11

Rationalize the denominator: $\frac{2}{\sqrt{5}}$.

We rationalize the denominator:

$$\frac{2}{\sqrt{5}} = \frac{2}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{5}.$$

If the denominator consists of multiple terms, one of which is a square root, we must rationalize it through another method: multiplying by conjugates.

Definition A1.1.12 (Conjugate)**.** For real numbers a and b and nonnegative number c, the conjugate of the expression $a + b\sqrt{c}$ is $a - b\sqrt{c}$.

We can see that multiplying $a + b\sqrt{c}$ by its conjugate rationalizes it:

$$(a + b\sqrt{c})(a - b\sqrt{c}) = a^2 - b^2c.$$

Example 13

Simplify the following:

$$\frac{3}{4 - \sqrt{7}}$$

We rationalize the denominator:

$$\frac{3}{4 - \sqrt{7}} \cdot \frac{4 + \sqrt{7}}{4 + \sqrt{7}} = \frac{3(4 + \sqrt{7})}{16 - 7} = \frac{3(4 + \sqrt{7})}{9} = \frac{4 + \sqrt{7}}{3}.$$

Defining Variables

An equation consists of an equal sign =, and two expressions that are on each side of the equal sign. Variables can be expressed in any symbol, such as shapes, but we usually use letters for convenience. Typically, we use them to represent unknown values in equations.

Combining Like Terms

The most basic technique when dealing with equations is combining like terms. This means combining all terms with the same variable at the same degree. For example, this means combining all terms with x or y^2.

Example 14

Simply $3x + 2x^2 + x - 3 + 7 + y - x^2$.

We combine like terms:

$$\begin{aligned} 3x + 2x^2 + x - 3 + 7 + y - x^2 &= (3x + x) + (2x^2 - x^2) + (-3 + 7) + y \\ &= 4x + x^2 + 4 + y \end{aligned}$$

Isolation Variables with One Variable Linear Equations

To solve for an unknown in an equation, we must isolate the variable. To do that, we can perform operations on both sides of the equation until the variable is isolated one side of the equation, and an expression made up of constants is produced. It is important to perform the same operations on both sides of the equation to keep the equation true.

Example 15
Solve for x: $\frac{x \cdot 6 \cdot 4}{3} - 5 = 2$.

We need to isolate the variable, x, so we first add 5 on each side, producing

$$\frac{x \cdot 6 \cdot 4}{3} = 7.$$

Then, we get rid of the coefficients surrounding x by dividing both sides by 6 and 4 and then multiplying both sides by 3:

$$\frac{x \cdot 6 \cdot 4}{3} \cdot \frac{3}{6 \cdot 4} = 7 \cdot \frac{3}{6 \cdot 4}$$
$$x = \frac{7}{8}$$

The previous example had only one solution. However, it is also possible for an equation to have no solutions or infinite solutions.

Example 16
Solve for x: $2x + 3 - x = \frac{12}{4} + x$.

First, we simplify the constants to get $2x + 3 - x = 3 + x$. Moving x to one side and constants to the other gives $2x - x - x = 3 - 3$. Simplifying both sides gives us $0 = 0$.

This expression is always true, so the equation has infinite solutions. We can check this by plugging in values for x to see if the equation is true.

Example 17
Solve for x: $3x - 3 = x + 4 - 2x$.

By moving x to one side and constants to the other, we get $3x - x + 2x = 4 + 3$. Simplifying the expressions on both sides of the equation gives $0 = 7$.

This is not true, so there are no solutions to this equation.

Systems of Linear Equations

A system of equations is one or more equations with one or more shared variables between them. The goal is to be able to solve for all the variables. Typically, a system of equations is solvable if there are the same amount of equations as variables.

Example 18

Solve the system of equations:

$$3x + 8 = y,$$
$$4 + 2y = 16x$$

We notice that in the first equation, y is isolated on the right side. Thus, we can use the expression $3x + 8$ to represent y.

We can replace y with the equivalent expression in the second equation:

$$4 + 2(3x + 8) = 16x.$$

We can then solve the second equation for x:

$$4 + 6x + 16 = 16x \Rightarrow 20 = 10x \Rightarrow 2 = x \Rightarrow x = 2.$$

Now that we have a value for x, we can plug it into the first equation to get the value of y:

$$3(2) + 8 = y \Rightarrow y = 14.$$

Our solution is $(x, y) = (2, 14)$.

Example 19

Solve the system of equations:

$$3y - x = 10,$$
$$y + 2x = 4 - y$$

We can first simplify the second equation:

$$2y + 2x = 4 \Rightarrow y + x = 2.$$

We then add the two equations together by adding the left sides together and equating them to the right sides added together:

$$4y = 12.$$

Next, we can then solve for y:

$$y = 3.$$

We plug this back into the simplified second equation to solve for x:

$$3 + x = 2 \Rightarrow x = -1.$$

Our solution is $(x, y) = (-1, 3)$.

Factoring and Distributing

With the distributive property, we can distribute multiplication across addition and subtraction. This can help when variables are within parentheses being multiplied.

Example 20

Solve for x: $3(x + 2) = 15$.

We apply the the distributive property:

$$3x + 6 = 15 \Rightarrow 3x = 9 \Rightarrow x = 3$$

We can also do the reverse, where instead of distributing, we group terms and then multiply them.

Example 21

Solve for x: $48x + 36 = 24$.

We notice that both 48 and 36 are multiples of 12, so we factor:

$$12(4x + 3) = 24.$$

This helps us simplify the equation by dividing:

$$4x + 3 = 2 \Rightarrow 4x = -1 \Rightarrow x = -\frac{1}{4}.$$

Simon's Favorite Factoring Trick

Example 22

Find all pairs of positive integers (x, y) such that $xy - 5x - 3y = 81$.

We can factor the first two terms on the left side, giving $x(y - 5) - 3y = 81$.

We would like to produce another $(y - 5)$ from the $3y$ so that we can group the left side into one product. We notice that we can do this by adding 15 on both sides and factoring:

$$x(y - 5) - 3y + 15 = 81 + 15 \Rightarrow x(y - 5) - 3(y - 5) = 96 \Rightarrow (x - 3)(y - 5) = 96.$$

We know that since x and y are both integers, $x - 3$ and $y - 5$ are also both integers, which means $x - 3$ and $x - 5$ are the factors of 96. Then, we can solve for all the pairs (x, y) by going through factors of 96.

For x and y to be positive, we must have $x - 3 \geq -3$ and $y - 5 \geq -5$.

We thus find the factor pairs of 96: $(1, 96)$, $(2, 48)$, $(3, 32)$, $(4, 24)$, $(6, 16)$, $(8, 12)$, $(12, 8)$, $(16, 6)$, $(24, 4)$, $(32, 3)$, $(48, 2)$, and $(96, 1)$, which directly map to the solutions after adding three to the first term and five to the second.

Our solutions are then $(4, 101)$, $(5, 53)$, $(6, 37)$, $(7, 29)$, $(9, 21)$, $(11, 17)$, $(15, 13)$, $(19, 11)$, $(27, 9)$, $(35, 8)$, $(51, 7)$, and $(99, 6)$.

This method is called Simon's Favorite Factoring Trick, which is used to cleverly solve Diophantine equations.

Squares of Binomials

Binomials are expressions with two terms, such as $5x + 13y$. The important thing to know is:

$$(a + b)^2 = a^2 + 2ab + b^2.$$

Important Factorizations

$$
\begin{aligned}
a^2 - b^2 &= (a-b)(a+b) && (6)\\
a^3 + b^3 &= (a+b)(a^2 - ab + b^2) && (7)\\
a^3 - b^3 &= (a-b)(a^2 + ab + b^2) && (8)\\
a^n - b^n &= (a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \cdots + ab^{n-2} + b^{n-1}) && (9)\\
a^n + b^n &= (a+b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \cdots - ab^{n-2} + b^{n-1}) && (10)\\
a^3 + b^3 + c^3 - 3abc &= (a+b+c)(a^2 + b^2 + c^3 - ab - bc - ac) && (11)
\end{aligned}
$$

Note that $a^n + b^n$ can only be factored if n is odd. Therefore, $a^2 + b^2$ cannot be factored.

Claim A1.1.23 (Sophie-Germain Identity) — The Sophie-Germain Identity states that

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2).$$